

Structure-Conditioned Pretraining for Genomic Sequence Models: A Timescale Ceiling and a Diagnosed Null

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Abstract

State-space sequence models are increasingly used as genomic foundation models because their linear cost admits contexts far beyond attention. We report two results about conditioning such a model on measured three-dimensional chromatin structure during pretraining, rather than attaching structure to frozen embeddings after the fact.

First, we identify a *timescale ceiling* that is present before training begins. In a selective state-space model the retention time obeys $\tau = 1/(\Delta|A| + p)$, so the initialisation constants of Δ and of any additive gate impose a hard upper bound on memory. With Mamba’s reference $\Delta_{\min} = 10^{-3}$ this bound is 10^3 tokens, which at single-nucleotide tokenisation is shorter than a typical gene; we measure 999.5. The ceiling is invisible to the standard diagnostic, which omits p . Correcting three constants at *zero* parameter cost moves median trained τ from 14.2 to 434.7 and takes the fraction of layer-directions reaching topological-domain scale from 0/32 to 32/32.

Second, with that ceiling removed, we find that conditioning Δ on Hi-C-derived structure gives *no* improvement in masked-language-modelling loss at matched parameters and matched compute (+0.0020 bits, 0.80σ , exact permutation $p = 0.600$), and we diagnose why. The structural signal is overwhelmingly *between* windows rather than within them: only 11.0% of its variance lies inside a 65,536 bp window. We show this is a property of Hi-C spatial autocorrelation and not of binning or window choice, by measuring it three ways — across window widths from 8 kb to 524 kb, across bin resolutions from 5 kb to 1 kb, and through the structural encoder. A per-position conditioning mechanism therefore addresses a minority of the available signal by construction, at every resolution and context length reachable on commodity hardware.

1 Introduction

The genome is read as a one-dimensional string but functions as a folded three-dimensional object. Regulatory contacts routinely join loci hundreds of kilobases apart in sequence that are adjacent in space, and this folding is measurable: chromosome conformation capture (Hi-C) yields contact maps from which topologically associating domains (TADs) and A/B compartments can be derived [4, 6].

Sequence models for genomics have grown contexts aggressively, first with efficient attention and more recently with state-space models (SSMs) whose cost is linear in length [2, 5, 7]. A natural question follows: if a model must learn long-range dependence, and that dependence is partly explained by measured folding, does supplying the folding help?

Existing work answers this by *post-hoc attachment*. Structure is applied to representations that are already frozen, typically through graph attention over a contact graph. This tests whether structure is useful for a downstream readout, but it cannot test whether structure changes what the encoder learns.

We ask the other question. We inject structure *during* pretraining, at the one place in a selective SSM where long-range behaviour is actually decided: the timescale Δ , which sets how long the recurrent state retains information. The hypothesis is direct — a model told where domain boundaries are should learn to reset at them and retain within them.

Answering it required first fixing something we did not expect to find. Our contributions are:

- **A timescale ceiling in SSM initialisation** (§4). The reference Δ initialisation caps retention at $1/\Delta_{\min}$ tokens before the first gradient step. The fix costs no parameters. We further show the ceiling recurs through a second constant — an additive gate term p omitted from the usual τ estimator — which silently reimposes a 55-token bound on the conditioned model while leaving the unconditioned baseline untouched.
- **A mechanism for structure-conditioned pretraining** (§5), costing +0.43% parameters and numerically identical to its baseline at initialisation, so that anything observed is learned.
- **A matched-parameter, matched-compute null, with controls** (§7). Loss is unchanged; input-substitution controls show the structural channel reaches the output but carries $\sim 0.05\%$ of the divergence that masking does.
- **A measured explanation** (§8). The conditioning signal is nearly constant inside the window it modulates, and we show this is intrinsic to Hi-C rather than an artefact of binning or context length.

We report the null as a result. The design was pre-registered, the effect-size floor was fixed before any number existed, and no threshold was moved afterwards.

2 Related work

Long-context genomic sequence models. Attention-based genomic models are limited by quadratic cost, which at single-nucleotide tokenisation binds quickly. Implicit-convolution and state-space architectures relaxed this: HyenaDNA [5] reaches 10^6 -token contexts at single-nucleotide resolution, and Caduceus [7] adds reverse-complement equivariance to a bidirectional Mamba backbone [2]. Our backbone is deliberately of this family and parameter-matched to the smallest published Caduceus configuration, so that the mechanism rather than the scale is what differs. The timescale ceiling of §4 is a property of the shared initialisation and therefore applies to this family generally, not only to the model studied here.

Structure-aware genomic models. A second line predicts folding *from* sequence: Akita and Orca learn contact maps directly, establishing that sequence carries substantial information about three-dimensional organisation. The converse direction — using measured folding to improve sequence representations — has been approached by attaching structure to representations that are already trained. Graph-attention methods of this kind operate over a contact graph on top of *frozen* embeddings. This is an effective way to exploit structure at readout time, but it leaves the encoder unchanged and therefore cannot answer whether structure alters what is learned. Our design differs in exactly this respect: conditioning is applied during pretraining and is removed at transfer time, so any benefit must reside in the learned sequence weights.

Conditioning recurrent timescales. Gating a recurrent model’s retention on side information has precedent in general sequence modelling, where forget-gate biases control effective memory length. What is specific here is the interpretation: in a selective SSM the timescale is an explicit, measurable quantity in tokens, so an external signal that predicts *where dependencies end* — which is what an insulation profile is — can be applied to it directly and its effect audited. That auditability is what makes the negative result of §7 diagnosable rather than merely disappointing.

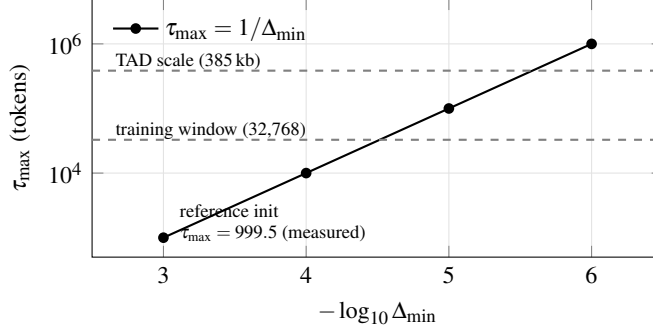


Figure 1: The ceiling of Eq. (2). Retention time cannot exceed $1/\Delta_{\min}$ regardless of training. At the reference $\Delta_{\min} = 10^{-3}$ the bound sits two orders of magnitude below the training window and three below domain scale.

3 Background

Selective state-space models. A selective SSM maintains a state h_t updated as $h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t$, where the discretisation $\bar{A}_t = \exp(\Delta_t A)$ makes the effective retention time input dependent. With A diagonal and negative, the characteristic decay time of direction i is

$$\tau_{t,i} = \frac{1}{\Delta_t |A_i|}, \quad (1)$$

so Δ is the quantity that decides how far back the model can see. In Mamba, $\Delta_t = \text{softplus}(W_{\Delta} x_t + b_{\Delta})$, and b_{Δ} is initialised so that Δ spans $[\Delta_{\min}, \Delta_{\max}]$.

Structural features. From a Hi-C contact matrix at 5 kb resolution we derive an eight-dimensional per-bin vector ϕ : insulation at three scales (100, 250, 500 kb), a 2 Mb directionality index, log contact density, upstream mass fraction, short-to-long range ratio, and the first principal component of the correlation matrix (the A/B compartment eigenvector). Validated against independently processed tracks, our insulation attains Pearson $r = 0.9969$ and our compartment eigenvector $r = 0.9759$.

Under reverse complementation the features carry signs $[1, 1, 1, -1, 1, -1, 1, 1]$; directionality and upstream mass fraction are antisymmetric. This flip must be applied to raw ϕ before encoding, since an encoder mixes the coordinates and destroys the correspondence.

4 A timescale ceiling before the first gradient step

Equation (1) has an immediate consequence that we have not seen stated. If $A_i = -(i+1)$ for $i = 0, \dots, d_{\text{state}} - 1$, as in the reference initialisation, then $|A_i| \geq 1$ and therefore

$$\tau_{\max} = \max_{t,i} \frac{1}{\Delta_t |A_i|} \leq \frac{1}{\Delta_{\min}}. \quad (2)$$

The bound involves no learned quantity. It is a property of the initialisation, present before any data is seen, and no amount of training can exceed it while Δ remains bounded below.

With Mamba’s reference $\Delta_{\min} = 10^{-3}$ this is 1,000 tokens. At one token per base pair — the tokenisation used by recent genomic SSMs — that is shorter than a typical human gene and two orders of magnitude below a TAD. We measure $\tau_{\max} = 999.5$ at initialisation, matching Eq. (2).

Table 1: Effect of the initialisation fix. Three seeds per row, 2,000 steps, 32,768 bp window. Parameter count is identical across rows.

	median trained τ	directions reaching TAD scale	val bits/nt
reference ($\Delta_{\min} = 10^{-3}$)	14.2	0/32	1.5210 ± 0.0040
corrected ($\Delta_{\min} = 10^{-6}$)	434.7	32/32	1.5197 ± 0.0025

The fix, and what it costs. Lowering $\Delta_{\min}$ to 10^{-6} raises the bound to 10^6 tokens. This changes three constants and *no parameters*. Table 1 reports trained models before and after.

A second constant, and why the diagnostic missed it. Our conditioned model (§5) adds a permeability gate p_t , giving $\tau = 1/(\Delta|A| + p)$. A floor on p is a ceiling on τ exactly as $\Delta_{\min}$ is, and with the gate bias initialised at -4 we have $p = 0.0182$ and hence $\tau \leq 1/p \approx 55$ tokens — *below* the uncorrected reference bound, and applying only to the conditioned arm.

This went undetected because the standard estimator computes τ from Δ alone, omitting p . Under that estimator the conditioned model reported a median τ of 487.3 against the baseline’s 434.7, i.e. an apparent *lengthening*; recomputed with p included, its true median is ≈ 51 tokens, roughly $8.5\times$ *shorter*. The quantity being reported was not the quantity governing memory.

We state this as a general caution. In any selective SSM, every additive term entering the discretisation imposes its own reciprocal ceiling, and a τ diagnostic that omits one will report a plausible number that is not the model’s memory. We found two such constants in one architecture.

5 Structure-conditioned pretraining

The backbone is a bidirectional Mamba masked language model: $d_{\text{model}} = 256$, 16 layers, $d_{\text{state}} = 16$, expansion 2, one token per base pair, trained to reconstruct masked nucleotides. The uniform-random loss floor is exactly 2.000 bits/nt.

Structural conditioning adds a pointwise encoder $E : \mathbb{R}^8 \rightarrow \mathbb{R}^2$ and, per layer, a matrix W_{struct} whose output is added to the pre-activation of Δ (Fig. 2):

$$\Delta_t = \text{softplus}(W_{\Delta}u_t + W_{\text{struct}}s_t + b_{\Delta}), \quad s_t = E(\phi_t). \quad (3)$$

Two design choices make the comparison interpretable. W_{struct} is *zero-initialised*, so at step 0 the conditioned model computes exactly the baseline function and any divergence is learned. And the two arms are *paired*: a reference baseline is constructed under the same seed and copied into every shape-matching tensor, giving 275/275 bitwise-identical parameters at initialisation, so seed variance is shared rather than confounded with arm.

Parameter counts are 7,725,312 (baseline) and 7,758,354 (conditioned), a $+0.43\%$ difference against a 5% budget.

6 Experimental setup

Data. Human chromosome 9, GM12878, Hi-C from a published deeply-sequenced experiment set, mapped to GRCh38. Of 27,679 bins at 5 kb, 21,519 (77.7%) survive coverage filtering. Windows of 32,768 bp; 5,422 train, 274 validation, 243 test.

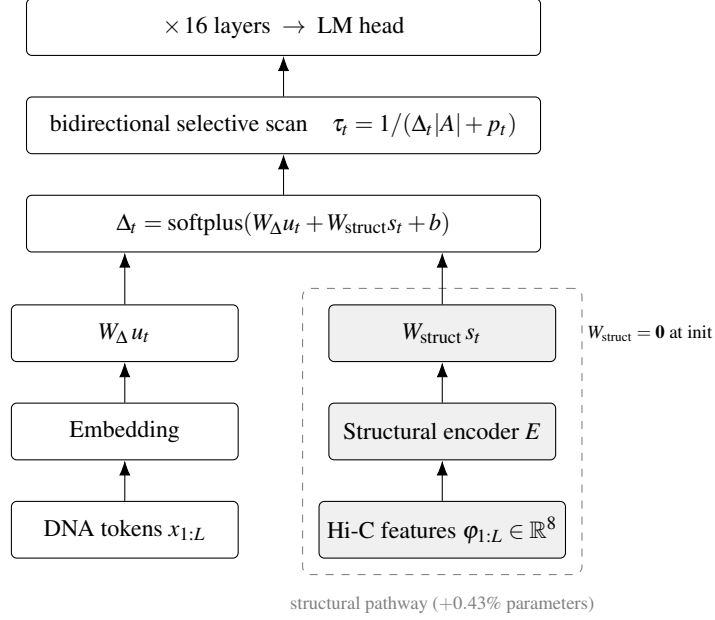


Figure 2: Structure enters at the timescale. The structural pathway (dashed) maps eight Hi-C features to a two-dimensional code and adds a learned bias to Δ_t inside the softplus, so it modulates *retention* rather than content. Because W_{struct} is zero-initialised, the model is numerically identical to the sequence-only baseline at step 0.

Table 2: Measured memory and throughput, single device, fp32, batch 2, with gradient checkpointing. Single process without gradient synchronisation; these are therefore a lower bound on distributed step time.

window (bp)	baseline s/step	conditioned s/step	peak memory (GiB)
32,768	3.02	3.43	3.81
65,536	6.28	7.17	7.51
131,072	15.45	16.98	14.90

Protocol. Three seeds per arm, 2,000 steps, fp32 throughout, matched on steps and tokens rather than wall clock. Across-seed standard deviation of the baseline is $\sigma = 0.0025$ bits, and the effect-size floor was fixed *in advance* at $2\sigma = 0.0050$ bits.

Controls. At evaluation only, on the trained conditioned model, we substitute ϕ with: s0 zeros; s1 a global permutation across bins; s2 a 10 Mb circular shift; s3 features recomputed from contacts resampled under a distance-matched null; s4 eight aligned one-dimensional sequence covariates. Controls draw from a fixed seed distinct from the training seed, so every training seed sees the same substituted structure.

Computational cost. Table 2 reports measured memory and step time, which bound what window widths are reachable and therefore what fraction of the structural signal is addressable at all (§8). Conditioning costs 10–14% in step time, from the two extra encoder passes required by the reverse-complement sign flip.

Table 3: Primary result. Matched parameters, matched compute, paired initialisation. Three seeds per arm.

	val bits/nt	median τ (reported)	per-seed
baseline	1.5197 ± 0.0025	434.7	1.5196 / 1.5172 / 1.5223
structure-conditioned	1.5217 ± 0.0045	487.3	1.5266 / 1.5177 / 1.5209
difference	$+0.0020$ (0.80σ)	permutation $p = 0.600$ (floor 0.100)	

Table 4: Input-substitution controls at evaluation. Masking floor 1.321×10^{-1} is context for scale, not a threshold.

control	Δ bits	symmetric KL	flip rate
s0 zeros	+0.0000	2.539×10^{-5}	0.0052
s1 permuted	+0.0001	6.334×10^{-5}	0.0073
s2 shifted	+0.0001	5.528×10^{-5}	0.0068
s3 distance-matched	+0.0001	6.827×10^{-5}	0.0075
s4 sequence-matched	+0.0000	2.874×10^{-5}	0.0052

7 Results

Language modelling is unchanged. Table 3 reports the primary comparison. The conditioned arm is 0.0020 bits *worse*, i.e. 0.80σ , well inside noise and with the sign pointing away from the hypothesis. The exact two-sided permutation test gives $p = 0.600$. We note explicitly that at three seeds per arm the minimum attainable p is 0.100, so significance was unreachable by design; this is a limitation of the pilot, not evidence of equivalence.

The channel is connected but carries almost nothing. Table 4 shows the input-substitution controls. Removing or corrupting ϕ changes the output measurably — the structural input does reach the prediction — but the induced divergence is $\sim 0.05\%$ of the divergence caused by masking itself. The ordering $s3 \approx s1 \approx s2 \gg s4 \approx s0$ is what one expects if divergence simply tracks the distance of substituted values from the true ones, s0 being the standardised mean. It shows the input arrives; it does not show that structure is used.

The pathway is trained but inert. Gradient pressure reached the structural pathway: the Frobenius norm of W_{struct} is non-zero in 32/32 layer-directions in every seed (median 0.19–0.22) against the exact zero it was initialised to. Yet its contribution to the variance of Δ is negligible. Defining $D_1 = \text{Var}_t(W_{\text{struct}}s_t) / \text{Var}_t(W_{\Delta}u_t)$ with a pre-registered inertness threshold of 0.05, we measure pooled medians of 0.0004, 0.0002, 0.0002 across seeds.

The permeability gate never engaged: initialised at $p = 0.018150$, its measured mean over 2.87×10^8 evaluations was 0.017234 / 0.017193 / 0.017108, with full range [0.0133, 0.0197] — all mass in a single bin.

One probe does separate the arms. Withholding ϕ at probe time and predicting within-window insulation from hidden states after removing window identity, every conditioned seed exceeds every baseline seed (conditioned +0.0254 / +0.0303 / +0.0255; baseline +0.0134 / +0.0146 / +0.0221), attaining the 3v3 permutation floor of $p = 0.100$. Two caveats travel with this. Both arms sit *below* a local GC-composition floor of +0.0802, so this is relative rather than absolute structural sensitivity — a contrast between two weak predictors. And two other targets give nothing.

Table 5: Frozen-probe accuracy on genomic classification tasks, three seeds per arm, against a k -mer composition floor ($k \leq 3$). Not comparable to published fine-tuned results. Both arms exceed the composition floor; the arms do not separate.

task	n_{test}	k -mer floor	baseline	conditioned	difference
mouse enhancers	242	0.7727	0.7741	0.7879	+0.0138
human enhancers (Cohn)	6,948	0.7083	0.7270	0.7185	−0.0084
human non-TATA promoters	9,034	0.8083	0.8413	0.8418	+0.0005

Neither arm separates on external benchmarks. Internal diagnostics cannot place a model relative to the literature, so we additionally evaluate on a standard suite of genomic classification tasks [3] under a *frozen-encoder linear probe*: the pretrained encoder is fixed, sequences are mean-pooled over positions, and a regularised logistic regression is fitted on the resulting representations. We emphasise that published numbers on these tasks are obtained by *fine-tuning* the whole model and are therefore not comparable to ours; a frozen probe is a lower bound on the same weights. What the protocol does support is a floor and the arm-versus-arm contrast, both arms receiving identical treatment.

Because these sequences have no associated Hi-C — and two tasks are not human — the conditioned arm is evaluated with ϕ set to zero. Its structural pathway is therefore inactive, and any difference reflects only what conditioning left in the sequence weights. This is the transfer setting that matters for a foundation model: at deployment, structure is usually unavailable.

Table 5 reports the completed subset. Both arms clear the composition floor on every task, by +0.019 and +0.033 on the two with large test sets — evidence that pretraining on a single chromosome produced representations exceeding trinucleotide composition. The arms do not separate: the mean difference is +0.0019 accuracy, and no task exceeds 1.6 standard errors. We note that the largest apparent gap, on mouse enhancers, occurs on the smallest test set, where the binomial standard error is 0.0266 and the observed difference is 0.52 of it. Across eight such tasks, one arm sweeping all three seeds of the other by chance has probability 0.57 under the null; we therefore do not interpret it.

8 Why: the conditioning signal is between windows, not within them

The mechanism biases Δ *per position*. It can therefore only exploit variation in ϕ that occurs *inside* a window. We quantify the available fraction as

$$\text{keep}(\phi) = \frac{\mathbb{E}_w [\text{Var}_{t \in w}(\phi_t)]}{\text{Var}_t(\phi_t)}. \quad (4)$$

On held-out windows $\text{keep}(\phi) = 0.0573$: 94% of the structural variance lies *between* windows. The mechanism’s input is nearly constant over the interval it modulates — by construction of the window size, not by any failure of training.

Two natural remedies fail, and we measured both rather than assuming (Fig. 3).

Widening the window is a partial mitigation, not a fix. Across widths from 8 kb to 524 kb, $\text{keep}(\phi)$ rises monotonically but slowly. At 65,536 bp — which we verify fits in 7.51 GiB with gradient checkpointing — 89.0% of the variance is still between windows. At 131,072 bp, which also fits (14.90 GiB) at $2.41 \times$ the step cost, 78.8% remains. Even at 524,288 bp, beyond practical memory, half of it does.

Refining the bins does not fix it either. We fetched a 10 Mb tile at 1 kb and 2 kb and recomputed ϕ with the same pipeline, against a same-tile 5 kb control. Like-for-like, 5 kb $\rightarrow$ 1 kb raises $\text{keep}(\phi)$ at

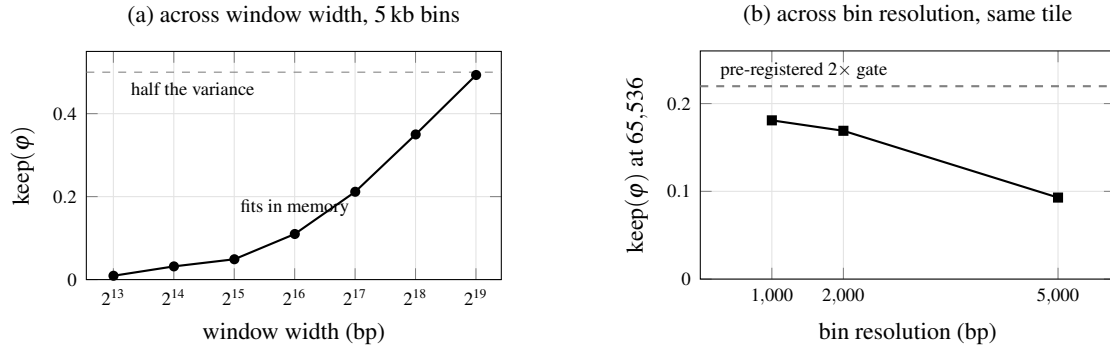


Figure 3: The signal is between windows at every reachable setting. (a) Widening the context raises $\text{keep}(\varphi)$ but even at 524,288 bp — far beyond what fits in memory — half the variance is still between windows. (b) Refining bins five-fold on a fixed tile raises it by $1.95\times$, missing a gate fixed in advance at $2\times$; at 1 kb, 81.9% of the variance remains between windows.

65,536 bp by $1.95\times$ ($0.0929 \rightarrow 0.1809$), missing a gate fixed in advance at $2\times$. Most of the gain arrives by 2 kb: the final step to 1 kb buys $1.071\times$ for $1.70\times$ the contact entries. At 1 kb bins — five times finer than any resolution used here — 81.9% of the variance is *still* between windows.

The encoder is not the bottleneck. One might suspect the two-dimensional structural code discards within-window variation. It does the opposite: the ratio of keep after encoding to keep of raw φ is $2.21 / 2.35 / 1.20$ across seeds, mean 1.92 . The encoder *increases* the within-window variance fraction of what passes through it.

Interpretation. Three independent interventions — context length, bin resolution, and the encoder — leave the conclusion unchanged. The dominant variation in Hi-C-derived features occurs on scales longer than any context these models can process, which is a statement about the spatial autocorrelation of chromatin contact maps rather than about one implementation. A per-position conditioning mechanism addresses the minority share by construction. This does not show that structure-conditioned pretraining cannot work; it shows that a per-position coupling is the wrong instrument, and that mechanisms operating at window or inter-window granularity are the ones with access to the remaining 80–90%.

9 Discussion and limitations

What we claim. That SSM timescale initialisation imposes ceilings that are invisible to the usual diagnostic, and that they are cheap to remove. That conditioning Δ on measured structure does not improve masked language modelling at this scale. And that the reason is measurable and architectural rather than incidental.

What we do not claim. We do not claim structure is useless for genomic representation learning. Our conditioned arm additionally paid the 55-token permeability ceiling of §4 that the baseline did not, so the reported null is partly a null about a handicapped mechanism; correcting this is the first item of ongoing work.

Limitations. Three seeds per arm places the minimum attainable permutation p at 0.100, so no significant positive was expressible. Validation loss was still descending at 2,000 steps, so this is a null from an undertrained model. Splits are internal to one chromosome, and because Hi-C features autocorrelate over megabases, held-out ϕ leaks more than held-out sequence — a bias *toward* the conditioned arm, which makes the null more rather than less credible but disqualifies any positive. And the scale is small: 7.7M parameters, one cell line, ≈ 0.5 B pretraining tokens. For calibration, a comparable published bidirectional-Mamba genomic model is parameter-matched to ours (7.73M, $d_{\text{model}} = 256$, 16 layers) but pretrained on roughly $100\times$ more tokens; we compare mechanism, not scale.

A caution for cross-cell-line evaluation. Evaluating structure-aware models across cell types is the natural next test, and a common design is to probe insulation on a held-out cell line. We advise against it, on measured grounds. TAD boundaries are substantially conserved across cell types, so such a probe can be passed by memorising the pretraining cell line. The usual remedy is to probe A/B compartments instead, on the premise that they are more cell-type-variable — but that premise is conditional on the pair of lines chosen. Computing all eight features on a common chromosome for three lines, we find compartment agreement of +0.8389 between GM12878 and K562 against insulation agreement of +0.7530: for this pair the compartment eigenvector is *more* conserved than insulation, inverting the premise. For GM12878 against IMR90 the premise holds (+0.6906 versus +0.7675). The distinction tracks lineage — GM12878 and K562 are both haematopoietic, IMR90 is fibroblast.

The implication is that a cross-cell-line probe should target the *differential* between lines rather than either feature in isolation: a model that memorised the pretraining line scores zero on a differential target by construction. We also recommend excluding contact-density features from such comparisons, since they show the lowest agreement across *every* pair including the two same-lineage lines — the signature of sequencing depth and copy-number differences rather than regulatory architecture.

Ongoing work. A leakage-safe split across seven chromosomes (train chr10–13, validate chr14–15, hold out chr9) is built and unused; five paired seeds per arm reduce the permutation floor to 0.008; and a window-level conditioning arm targets the between-window share identified in §8. Those results are not reported here.

10 Conclusion

Selective state-space models carry a memory horizon fixed at initialisation, and we found two independent constants that cap it — one in the reference initialisation of every Mamba-family model, one specific to an additive gate and hidden from the standard estimator. Removing the first costs no parameters and increases trained retention thirty-fold.

With that ceiling lifted, conditioning the timescale on measured chromatin structure left language-modelling loss unchanged. Rather than report that as an uninformative negative, we located its cause: the structural signal varies on scales longer than the model’s context, at every resolution and window width reachable on current hardware, so a per-position mechanism can address only a tenth of it. That finding constrains the design space for structure-aware genomic models more usefully than the null alone.

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